Lab#7: Interrupt Instructions (Part B)

Assignment #1: Write an assembly code to display 'Assembly' message, then moving down the message until reach the end of the screen .

Solution:

```
ORG 100H
mov BL,0
     label1:      mov AH,2H
                  mov BH,0
                  mov DH,BL
                  mov DL,BL
                  INT 10H
                  mov AH,9H
                  mov DX,offset st1
                  INT 21H
                  mov AH, 6H
                  mov AL, 0
                  mov BH, 7
                  mov CX, 0
                  mov DL, 79
                  mov DH, 24
                  int 10H
                  ADD BL,1
                  cmp BL,23
                  jz label2
                  loop label1
label2:           mov AH,2H
                  mov BH,0
                  mov DH,BL
                  mov DL,BL
                  INT 10H
                  mov AH,9H
                  mov DX,offset st1
                  INT 21H
                  mov AH, 6H
                  mov AL, 0
                  mov BH, 7
                  mov CX, 0
                  mov DL, 79
                  mov DH, 24
                  int 10H
                  SUB BL,1
                  cmp BL,1
                  jz label1
                  loop label2
                  mov AH,4CH
                  INT 21H
                  hlt
     st1 db "Assembly",13,10,'$'
```

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Assignment #3: This example prints out “hello world!" by writing directly to video memory.
In VGA memory: first byte is ASCII character, byte that follows is character attribute. If you change the second byte, you can change the color of the character even after it is printed. The character attribute is 8 bit value; high 4 bits set background color and low 4 bits set foreground color.

HEX	BIN	COLOR
0	0000	black
1	0001	blue
2	0010	green
3	0011	cyan
4	0100	red
5	0101	magenta
6	0110	brown
7	0111	light gray
8	1000	dark gray
9	1001	light blue
A	1010	light green
B	1011	light cyan
C	1100	light red
D	1101	light magenta
E	1110	yellow
F	1111	white

```
01  ORG 100H
02
03      MOV AX,3
04      INT 10H
05      MOV AX, 0B800H
06      MOV DS, AX
07
08      MOV [02h], 'H'
09      MOV [04h], 'e'
10      MOV [06h], 'l'
11      MOV [08h], 'l'
12      MOV [0ah], 'o'
13      MOV [0ch], ' '
14      MOV [0eh], 'W'
15      MOV [10h], 'o'
16      MOV [12h], 'r'
17      MOV [14h], 'l'
18      MOV [16h], 'd'
19      MOV [18h], '!'
20
21      MOV CX, 12
22      MOV DI, 03H
23  C:  MOV [DI], 11101100B
24      ADD DI, 2
25      LOOP C
26
27      MOV AH, 0
28      INT 16H
29
30  HLT
```

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